OZNAL Capstone Project - Analysis & Strategy

Assignment Summary

- **Course**: The Elements of Statistical Learning (Feb-Jun 2026, STU FIIT)
- **Goal**: Choose a real-world dataset, define a hypothesis, select 2 modeling scenarios, present in Shiny app + Jupyter notebook + one-pager
- **Key requirement**: Models must involve **several dozen parameters (ideally >30)**
- **Scenarios chosen**: Must pick 2 from list of 5
- **Deliverables**: One-pager (5pts), Project documentation (30pts), Shiny app (10pts), Free questions (15pts) = 60pts total, need 40 to pass
- **Tools**: Python/Jupyter or R/Quarto, Shiny app for presentation
- **Team**: 2 people

Dataset: Oracle's Elixir (LoL Esports)

Source

- **Website**: https://oracleselixir.com/tools/downloads
- **Data dictionary**: https://oracleselixir.com/match-data/match-data-dictionary/
- **Files**: 2023, 2024, 2025 yearly CSVs

Volume

Year	Total Rows	Games	Complete Games	Completeness
2023	133,428	11,119	9,378	84%
2024	122,388	10,199	8,808	86%
2025	120,636	10,053	9,236	92%
Total	**376,452**	**31,371**	**27,422**	**87%**

Structure

- **165 columns** per file (identical across years)
- **12 rows per game**: 10 player rows (5 per team) + 2 team summary rows
- **Positions**: top, jng, mid, bot, sup, team
- **Target**: `result` (binary: 1=win, 0=loss), perfectly balanced (50/50)
- **Filter**: Use `datacompleteness == 'complete'` only

Data Quality

- No structural missing values in complete data
- Timeline data (goldat10/15/20/25) available for ~84% of player rows
- Objective data (dragons, barons, etc.) available on team rows (~84-100%)
- Bans available for ~99% of team rows
- `void_grubs` only 7.3% available (new 2024 mechanic) - exclude
- `pick1-pick5` 100% missing on player rows (team rows only, partial) - exclude

---

Complete Feature Reference (165 columns)

1. Game Metadata (identifiers, not model features)

Feature	Description	Use
`gameid`	Unique game identifier	Join key

Feature	Description	Use
`datacompleteness`	"complete" or "partial"	Filter to complete
`url`	Match history link	Reference only
`league`	Competition name (LCK, LEC, LPL, LCS, etc.)	Categorical feature (48 leagues)
`year`	Season year	Temporal split
`split`	Season split (Spring/Summer)	Categorical
`playoffs`	Was this a playoff game?	Binary feature
`date`	Date of the match	Temporal ordering
`game`	Game number in best-of series (1, 2, 3…)	Ordinal
`patch`	Game version (e.g., 13.1) - affects champion balance	Categorical (21 patches/year)
`participantid`	Player slot 1-10 (1-5 blue, 6-10 red)	Maps to role
`side`	Blue or Red team	Binary (Blue has first pick)
`position`	top/jng/mid/bot/sup/team	Role identifier
`playername`	Player's in-game name	Identifier
`playerid`	Unique player ID	Identifier
`teamname`	Team organization name	Identifier (~500 teams)
`teamid`	Unique team ID	Identifier

2. Draft (Champion Select)

Feature	Description	Availability
`champion`	Champion played (163 unique)	100%
`ban1-ban5`	5 champions banned by this team	~99% (team rows)
`pick1-pick5`	Draft order picks	Mostly missing
`firstPick`	Team had first pick in draft	~78%

3. Core Combat Stats (per player, 100% available)

Feature	Description	Corr with win
`kills`	Enemy champions killed	0.362
`deaths`	Times died	-0.509
`assists`	Kill participations	0.535
`teamkills`	Team total kills	0.678
`teamdeaths`	Team total deaths	-0.676
`doublekills`	2 kills in rapid succession	0.273
`triplekills`	3 kills in rapid succession	0.167
`quadrakills`	4 kills in rapid succession	0.080
`pentakills`	All 5 enemies killed rapidly	0.042
`firstbloodkill`	Got the first kill (binary)	0.074
`firstbloodassist`	Assisted first kill (binary)	0.088
`firstbloodvictim`	Was first to die (binary)	-0.074
`team kpm`	Team kills per minute	0.684
`ckpm`	Combined kills per minute (both teams)	~0 (symmetric)

4. Economy / Gold (per player, 100% available)

Feature	Description	Corr with win
`totalgold`	Total gold earned (includes passive)	0.293
`earnedgold`	Gold earned through actions only	0.326
`earned gpm`	Earned gold per minute	0.387
`earnedgoldshare`	% of team's earned gold	Role-dependent
`goldspent`	Gold spent on items	0.167
`gspd`	Gold Spent Percentage Difference vs opponent	0.762 (team)
`gpr`	Gold Percent Rating	0.717 (team)

5. Damage (per player, ~100% available)

Feature	Description	Corr with win
`damagetochampions`	Total damage to enemy champions	0.171
`dpm`	Damage per minute	0.209
`damageshare`	% of team's total damage	Role-dependent
`damagetakenperminute`	Damage absorbed per minute	0.114
`damagemitigatedperminute`	Damage blocked/shielded per minute	~0.011
`damagetotowers`	Damage to enemy towers	0.396

6. Vision / Map Control (per player, ~100% available)

Feature	Description	Corr with win
`wardsplaced`	Vision wards placed on map	0.026
`wpm`	Wards placed per minute	0.031
`wardskilled`	Enemy wards destroyed	0.062
`wcpm`	Wards cleared per minute	0.073
`controlwardsbought`	Purchased control wards	0.036
`visionscore`	Riot's composite vision metric	0.091
`vspm`	Vision score per minute	0.109

7. Farming / Creep Score (per player, ~100% available)

Feature	Description	Corr with win
`total cs`	Total minions + monsters killed	0.049
`minionkills`	Lane minions killed	0.020
`monsterkills`	Jungle monsters killed	0.051
`monsterkillsownjungle`	Monsters in own jungle	84% avail
`monsterkillsenemyjungle`	Monsters in enemy jungle (invading)	84% avail
`cspm`	Creep score per minute	0.058

8. Objectives (team rows only, join via gameid+side)

Feature	Description	Corr with win (team)
`firstblood`	Team got first kill	0.124
`firstdragon`	Team killed first dragon	~0.30
`firstherald`	Team killed first Rift Herald	~0.25
`firstbaron`	Team killed first Baron Nashor	0.663
`firsttower`	Team destroyed first tower	~0.35
`firstmidtower`	Team destroyed first mid tower	~0.40

Feature	Description	Corr with win (team)
`firsttothreetowers`	Team first to destroy 3 towers	~0.50
`dragons` / `opp_dragons`	Dragons killed by team/opponent	0.615
`elementaldrakes` / `opp_elementaldrakes`	Elemental drakes	0.616
`infernals`	Infernal drakes (bonus AD/AP damage)	Part of elementaldrakes
`mountains`	Mountain drakes (bonus armor/MR)	Part of elementaldrakes
`clouds`	Cloud drakes (bonus movement speed)	Part of elementaldrakes
`oceans`	Ocean drakes (bonus health regen)	Part of elementaldrakes
`chemtechs`	Chemtech drakes (bonus tenacity/heal)	Part of elementaldrakes
`hextechs`	Hextech drakes (bonus ability haste)	Part of elementaldrakes
`elders` / `opp_elders`	Elder dragons (powerful late-game execute)	~0.40
`heralds` / `opp_heralds`	Rift Heralds (early tower push)	~0.25
`barons` / `opp_barons`	Baron Nashors (team combat buff)	0.639
`towers` / `opp_towers`	Towers destroyed	**0.890** (strongest!)
`turretplates` / `opp_turretplates`	Early turret plates (gold)	~0.30
`inhibitors` / `opp_inhibitors`	Inhibitors destroyed	0.753

9. Early-Game Timeline (per player, ~84% available)

Available at 10, 15, 20, and 25 minute marks:

Feature Pattern	Description	Corr with win (@15)
`goldat{10,15,20,25}`	Player gold at timestamp	0.171
`xpat{10,15,20,25}`	Player XP at timestamp	0.062
`csat{10,15,20,25}`	Creep score at timestamp	0.038
`opp_goldat{10,15,20,25}`	Lane opponent's gold	0.171
`opp_xpat{10,15,20,25}`	Lane opponent's XP	0.062
`opp_csat{10,15,20,25}`	Lane opponent's CS	0.038
`golddiffat{10,15,20,25}`	Gold lead vs lane opponent	**0.343**
`xpdiffat{10,15,20,25}`	XP lead vs lane opponent	**0.265**
`csdiffat{10,15,20,25}`	CS lead vs lane opponent	0.199
`killsat{10,15,20,25}`	Kills at timestamp	0.149
`assistsat{10,15,20,25}`	Assists at timestamp	0.178
`deathsat{10,15,20,25}`	Deaths at timestamp	-0.172

Hypothesis

"Can early-game performance (first 15 minutes) predict the outcome of a professional League of Legends match, and which early-game factors are most predictive across different model families?"

Why this hypothesis works:

1. **No data leakage** - early-game stats are mid-game snapshots, not post-game results
2. **Practically useful** - coaches could identify when games are "decided"
3. **Moderate correlations** (0.25-0.35) - models need to actually work, not trivially predict
4. **Well-supported by literature** - prior work achieves 74-85% accuracy
5. **Rich feature space** - gold/xp/cs diffs + first objectives + kills/deaths = easily 30+ features

Alternative framing (if needed):

- "Which objective priorities (first dragon vs first herald vs first tower vs first blood) most predict winning?"
- "How does the predictive power of early-game features vary across different competitive leagues (LCK vs LEC vs LPL)?"

Recommended Scenarios

Scenario 1: Classification Comparison (3 methods + 2 partitioning)

Methods: 1. **Logistic Regression** - linear, interpretable baseline; coefficients directly show feature importance 2. **Random Forest** - non-parametric ensemble; handles interactions, provides Gini importance 3. **XGBoost (Gradient Boosted Trees)** - state-of-the-art; handles non-linearity, built-in regularization

Partitioning approaches: 1. **Stratified k-fold CV** (k=5 or 10) - standard, maximizes training data 2. **Temporal split** - train on 2023-2024, test on 2025 (realistic deployment scenario, accounts for meta shifts)

What to compare: - Accuracy, AUC-ROC, Precision, Recall, F1 - Which features are top-10 in each model - Explainability: LR coefficients vs RF Gini importance vs XGBoost SHAP values - Overfitting: training vs validation performance gap

Scenario 3: Algorithmic vs Embedded Feature Selection

Algorithmic (wrapper) methods: 1. **Recursive Feature Elimination (RFE)** - iteratively removes least important features 2. **Forward selection** - starts empty, adds best feature one at a time

Embedded methods: 1. **Lasso (L1 regularization)** - zeros out unimportant coefficients 2. **Random Forest feature importance** - Gini impurity-based ranking 3. **XGBoost built-in importance** - gain/cover/frequency metrics

What to report: - How many features each method retains - Which features survive across ALL methods (robust predictors) - Performance curve: accuracy vs number of features - Which features lose statistical significance as you add/remove predictors

---

Project Steps

Step 1: Data Preparation

- Load 2023-2025 Oracle's Elixir CSVs
- Filter to `datacompleteness == 'complete'`
- Decide unit of analysis: **team-level per game** (merge 5 player rows into 1 team row + join objective data)
- Handle missing timeline data (~16% missing)
- Remove irrelevant columns (identifiers, URLs)

Step 2: Feature Engineering

- **Early-game team aggregates**: avg/sum of golddiffat15, xpdiffat15, csdiffat15 across 5 players
- **First objective flags**: firstblood, firstdragon, firstherald, firsttower
- **Role-specific features**: carry gold share, support vision, jungle invade ratio
- **Draft features**: champion encoding (frequency or one-hot)

- **Meta features**: side, playoffs, league tier
- Target: `result` (binary)
- Expected: **40-60 features** before champion encoding

Step 3: EDA (Targeted Visualizations)

- Correlation heatmap of early-game features vs result
- Win rate by first objective secured (bar chart)
- Gold diff at 15 distribution: wins vs losses (density plot)
- Feature importance preview (quick RF or mutual information)
- Timeline visualization: how prediction confidence grows from 10 to 25 min

Step 4: Scenario 1 - Classification Comparison

- Implement Logistic Regression, Random Forest, XGBoost
- Run both partitioning strategies (k-fold CV + temporal split)
- Generate: ROC curves, confusion matrices, classification reports
- Compare explainability across models
- Discuss trade-offs: accuracy vs interpretability

Step 5: Scenario 3 - Feature Selection

- Run RFE, forward selection, Lasso, RF importance, XGBoost importance
- Plot: accuracy vs number of features for each method
- Venn diagram or table: which features selected by which methods
- Identify "universal" predictors that survive all methods
- Discuss information loss at each reduction step

Step 6: Shiny App

- **Tab 1**: Data explorer (filter by league, patch, role, year)
- **Tab 2**: Model comparison (select model, adjust hyperparameters, see performance update)
- **Tab 3**: Feature selection (interactive feature importance, toggle features on/off)
- **Tab 4**: Predictions (input early-game stats, get win probability)

Step 7: Documentation & Deliverables

- **Jupyter Notebook**: Full reproducible workflow with explanations
- **One-pager**: Executive summary written manually (NO LLM text per assignment rules)
- **Shiny app**: For oral presentation defense

Related Academic Papers

Papers directly using Oracle's Elixir data:

1. **"Feature Analysis to League of Legends Victory"** (IEEE CoG 2021)
   - Data: Oracle's Elixir, 2,840 pro matches
   - Methods: Random Forest, SVM, Logistic Regression
   - Finding: RF, SVM, LR best; Gini impurity for feature ranking
   - PDF: https://ieee-cog.org/2021/assets/papers/paper_292.pdf
2. **"Exploring the Most Important Variables for Winning"** (Tilburg University Thesis)
   - Data: Oracle's Elixir, Jan-Sep 2020
   - Methods: Random Forest, SVM, kNN
   - PDF: http://arno.uvt.nl/show.cgi?fid=156282
3. **Early Game Prediction Project** (GitHub - jadenoca)
   - Data: Oracle's Elixir 2021
   - Focus: Early game stats for outcome prediction

- Repo: https://github.com/jadenoca/LolEsportsData

Papers using similar LoL esports data:

4. **"League of Legends: Real-Time Result Prediction"** (arXiv 2023)
   - Methods: Linear Regression, LSTM
   - Finding: LSTM 93% accuracy; LR 82%
   - PDF: https://arxiv.org/pdf/2309.02449
5. **"The SIDO Performance Model for League of Legends"** (arXiv 2024)
   - Data: 10,390 games (2023), tested on 9,670 (2024)
   - Methods: Logistic Regression, MLP, SVM, XGBoost (13 features)
   - Focus: Role-sensitive framework, Victory Contribution metric
   - PDF: https://arxiv.org/pdf/2403.04873
6. **"Machine Learning Methods for Predicting LoL Game Outcome"** (IEEE TG 2022)
   - Finding: 74.4% accuracy from early-game data with probabilistic SVM
   - Link: https://ieeexplore.ieee.org/document/9720122/
7. **"E-Sports Player Performance Metrics Considering Player Roles"** (Springer 2022)
   - Focus: Role-based features affect prediction differently
   - Link: https://link.springer.com/article/10.1007/s42979-022-01660-6
8. **"Deep Learning Techniques for Identifying KPIs in LoL"** (ScienceDirect 2025)
   - Focus: Win prediction, map navigation, vision control as KPIs
   - Link: https://www.sciencedirect.com/science/article/pii/S2451958825001332
9. **"Explainable E-sports Win Prediction through ML Classification"** (arXiv 2025)
   - Focus: Explainability of predictions
   - Link: https://arxiv.org/html/2510.19671v1

Key findings from literature:

Finding	Accuracy	Implication
Early game (10-15 min) prediction	74-85%	Achievable, non-trivial
Post-game prediction	90-95%	Too easy / data leakage
RF, XGBoost, LR are standard	-	Perfect for Scenario 1
Feature selection heavily studied	-	Perfect for Scenario 3
Most papers use 2020-2022 data	-	Our 2023-2025 data adds novelty
Role-based analysis adds value	-	Differentiation opportunity

Data Sufficiency Assessment

VERDICT: Data is more than sufficient.

Requirement	Status
Several dozen parameters (>30)	40-60 features easily achievable, 200+ with champion encoding
Raw data, not pre-solved	Oracle's Elixir is raw match data, not a Kaggle competition
Enough volume for ML	27,422 complete games = ~274K player rows
Balanced target	50/50 win/loss
Supports 2 scenarios	Scenario 1 (classification comparison) + Scenario 3 (feature selection) fully supported
Shiny app viable	Rich filtering dimensions (league, patch, role, year)

Requirement	Status
Timeline for temporal split	3 years (2023-2025) enables train/test by year

Important caveat:

Post-game stats (kills, gold, towers) are **strongly correlated with winning by definition** (data leakage). Frame the project around **early-game prediction** (15-min snapshot) to avoid this. Acknowledge in the report.

2025 Deep Dive - Key Statistics & Findings

Dataset (2025 only, complete games)

- **9,236 games**, 92,360 player rows, 18,472 team rows
- **44 leagues**, 443 teams, 2,780 players, 172 champions
- **24 patches** (15.01 to 15.24)
- Date range: Jan 11 - Dec 29, 2025
- Average game length: **32.0 min** (median 30.9, std 5.5)

2025-Specific Mechanics

Atakhan (new 2025 objective, 99.7% data available): - Team that got Atakhan: **79.6% win rate** (n=8,947) - Team that didn't: **22.0% win rate** (n=9,473) - Only 2.9% of games had no Atakhan claimed at all - **Very strong predictor** - comparable to Baron in impact

Void Grubs (100% data available in 2025): - 0 grubs: 36.4% WR | 3 grubs: 57.5% WR | 6 grubs (all): 69.9% WR - Monotonic relationship with win rate (slight dip at 4 grubs) - Getting the full set of 6 gives ~34% win rate boost over getting 0

Dragon Soul progression: - 0 dragons: 3.6% | 1: 18.4% | 2: 46.4% | 3: 69.7% | **4 (soul): 89.9%** | 5: 94.9% - The soul threshold (4 drakes) is the critical tipping point

Elder Dragon: Got elder -> 82.1% WR (n=792)

Comeback Analysis

Gold deficit per player @15min	Win rate	Sample
Behind by >3000g	0.0%	n=12
Behind by >2000g	0.0%	n=75
Behind by >1500g	0.0%	n=87
Behind by >1000g	1.7%	n=542
Behind by >500g	8.7%	n=2,694
Any deficit	27.1%	n=9,235

Key insight: In 2025 pro play, a 1000+ gold per player lead at 15 minutes is essentially game over (98.3% win rate for the leading team). Comebacks only happen from small deficits.

Snowball Chains (First Objective Combinations)

Objectives secured	Win Rate	n
All 4 (FB+FD+FH+FT)	**85.8%**	2,006
FD+FH+FT	81%	1,149
FB+FH+FT	74%	1,600
FH+FT (Herald + Tower)	68%	1,223

Objectives secured	Win Rate	n
FB only	19%	1,159
None of the 4	**14.2%**	2,014

Key insight: First Herald + First Tower is the strongest 2-objective combo (68%). First Blood alone is nearly worthless (19%) - barely better than getting nothing. Tower-related firsts dominate.

First Blood by Role

Who gets the kill: - JNG: 29.3% | BOT: 26.6% | MID: 18.5% | TOP: 16.0% | SUP: 9.5%

Who dies first: - SUP: 23.9% | TOP: 23.3% | MID: 19.3% | JNG: 17.2% | BOT: 16.3%

Insight: Junglers are the primary first blood playmakers. Supports and top laners are the most vulnerable. BOT rarely dies first despite being in a 2v2 lane.

Game Pace by League (2025)

League	Avg Game Length	Team KPM	Style
PCS	30.1 min	0.65	Bloodiest, fastest
LAS	30.3 min	0.62	Aggressive
LJL	30.2 min	0.60	Aggressive
LCK	32.3 min	0.45	Methodical, low-kill
LEC	33.6 min	0.40	Slowest, most controlled
LFL	34.0 min	0.43	Longest games

Insight: Massive style differences. PCS has 63% more kills per minute than LEC. This means league should be a strong feature - models may need to account for regional playstyle.

Major vs Minor League

	Game Length	Team Kills	Games
Major (LCK/LEC/LPL/LCS)	32.7 min	14.1	861
Minor (all others)	31.9 min	16.6	8,375

Major leagues play slower with fewer kills. Minor leagues are more chaotic.

Playoffs vs Regular Season

Almost no statistical difference in 2025: - Kills, deaths, DPM, CSPM, VSPM, GPM all within 1% of regular season values - Playoffs are slightly longer (32.3 vs 31.9 min) and slightly less kill-heavy (0.509 vs 0.522 kpm)

Side Advantage (2025)

Blue side + first pick: **53.3% WR** (n=9,225). Consistent ~3% advantage.

Champion Meta (2025)

Most played: Rell (2,792), Xin Zhao (2,450), Varus (2,253), Corki (2,225), Ambessa (2,218)

Highest WR (100+ games): Poppy (55.4%), Taliyah (55.2%), Xayah (54.4%), Rumble (54.0%)

Lowest WR (100+ games): Seraphine (40.3%), Kennen (41.0%), Milio (43.0%), Mel (43.9%), Ornn (43.9%)

Most banned: Varus (37.4%), Vi (35.3%), Gwen (34.8%), Azir (30.9%), Rumble (30.8%)

Notable flex picks: Ambessa (top/mid/jng), Aurora (mid/top), Corki (bot/mid), Galio (mid/sup/top), Gwen (jng/top)

Early Game Prediction Power (2025 team-aggregated)

Feature	Correlation with Win
avg_golddiff15	+0.523
avg_xpdiff15	+0.492
avg_csdiff15	+0.449
firsttower	+0.402
firstherald	+0.329
sum_killsat15	+0.271
firstblood	+0.155
firstdragon	+0.152

Simple "gold lead at 15 -> win" rule: 72.9% accuracy

Post-Game Feature Correlations (2025 team-level)

Feature	Correlation
towers	+0.892
inhibitors	+0.755
teamkills	+0.701
dragons	+0.595
atakhans	+0.576
barons	+0.555
firstbaron	+0.593
firsttower	+0.402
firstherald	+0.327
void_grubs	+0.193
firstblood	+0.155
firstdragon	+0.152
teamdeaths	-0.700

Exploratory Visualizations

All plots saved in `visualizations/` folder. Generated from 27,422 complete games across 2023-2025.

Plot 1: Win Rate by First Objective (`01_first_objective_winrates.png`)

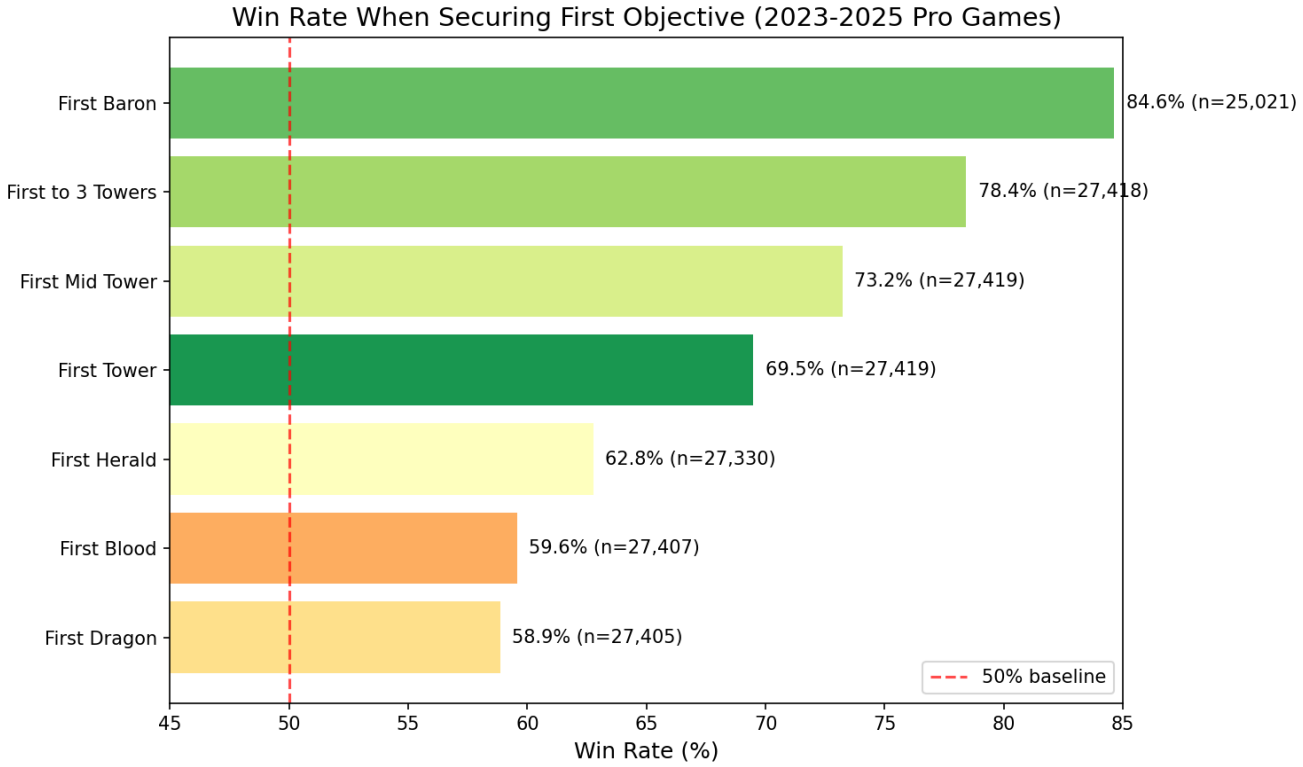

Figure 1: First Objective Win Rates

Key insights: - **First Baron** is the strongest single predictor: 84.6% win rate when secured - **First to 3 Towers** (78.4%) and **First Mid Tower** (73.2%) are strong mid-game indicators - **First Tower** overall: 69.5% win rate - **First Blood** (59.6%) and **First Dragon** (58.9%) have the weakest individual predictive power - **First Herald** (62.8%) sits in between - slightly more impactful than first blood - Implication: structural objectives (towers, baron) matter far more than combat-first (first blood) or early neutral objectives (dragon)

Plot 2: Gold Diff @15min Distribution (`02_golddiff15_distribution.png`)

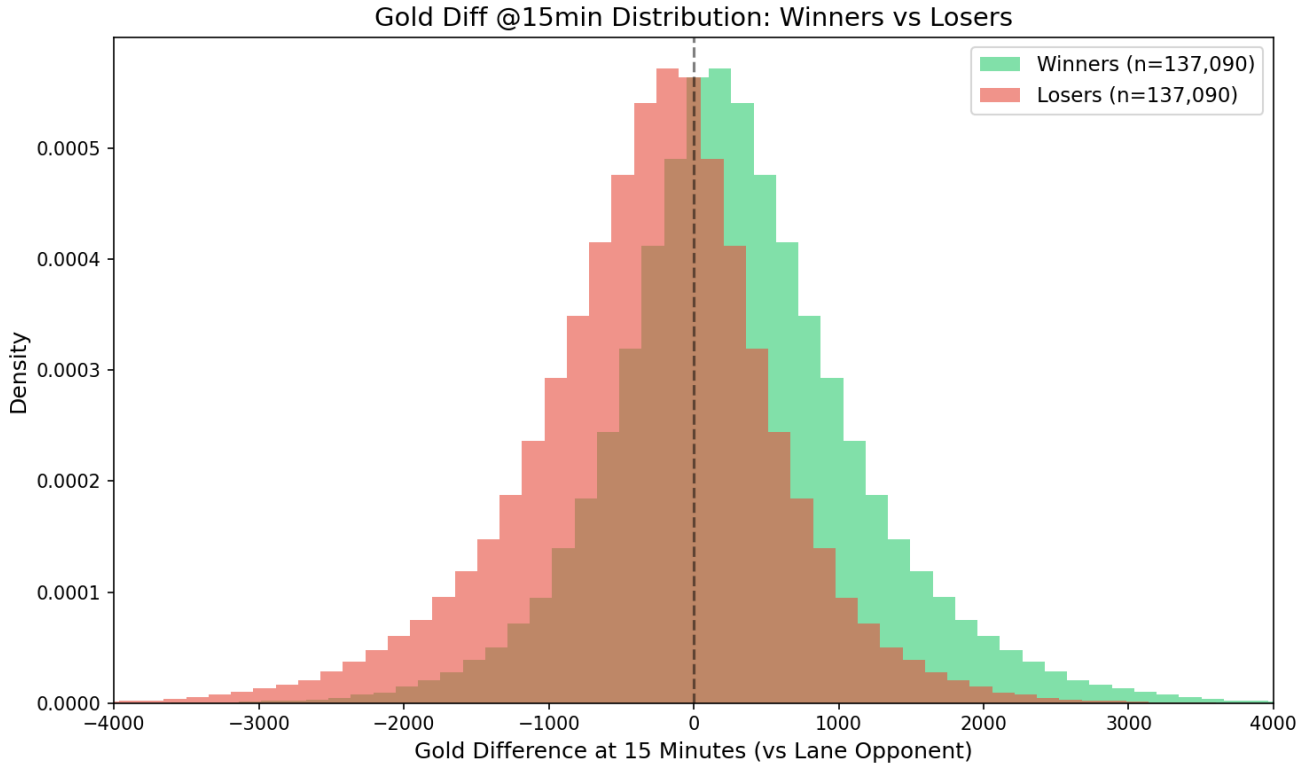

Figure 2: Gold Diff Distribution

Key insights: - Clear separation between winners (shifted right) and losers (shifted left) - However, there is **massive overlap** near zero - many games are still close at 15 minutes - Winners have a median gold lead of ~300-500 gold per player at 15 min - The tails diverge more strongly: large gold leads (>1500) are very predictive - This confirms the hypothesis is non-trivial: early-game data alone won't give 95% accuracy, which is good - models actually need to work

Plot 3: Predictive Power Growth Over Time (`03_prediction_power_over_time.png`)

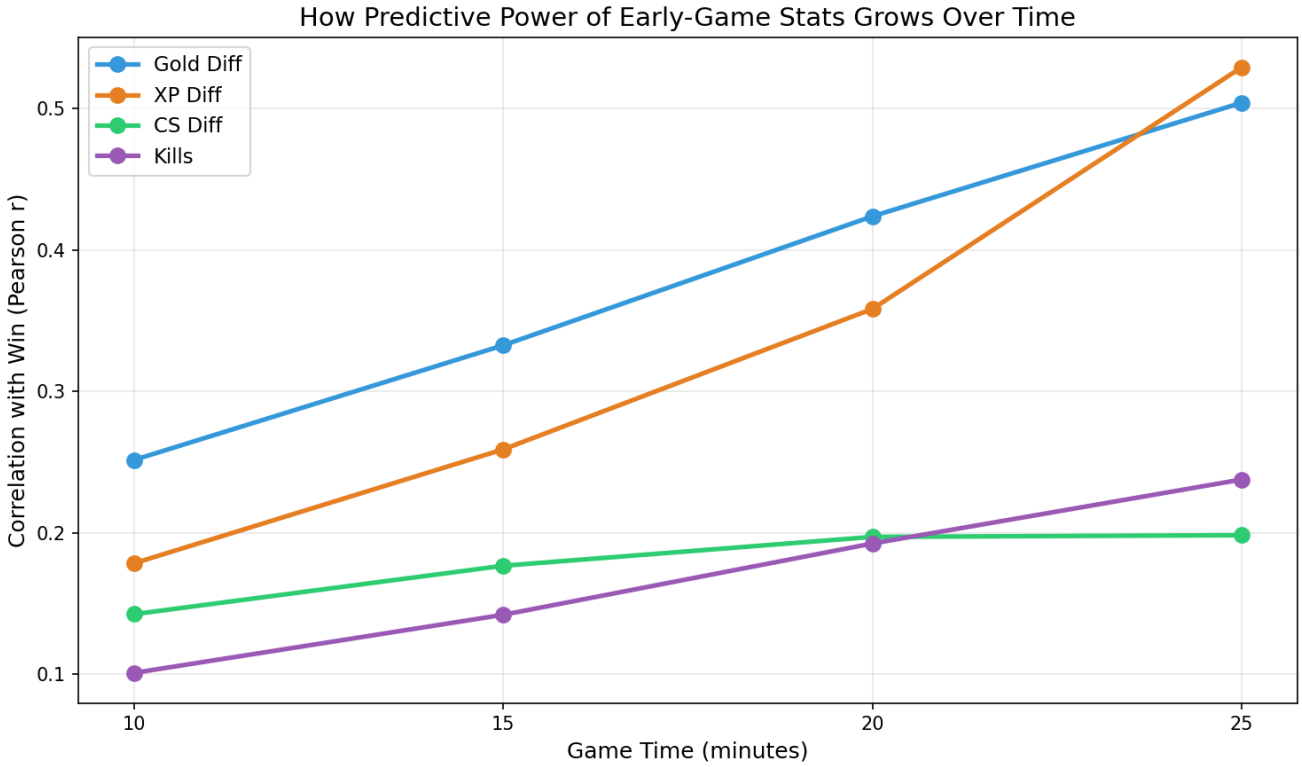

Figure 3: Prediction Power Over Time

Key insights: - **Gold Diff** is the most predictive early metric (r=0.25 at 10min, growing to 0.50 at 25min) - **XP Diff** starts weaker but catches up and **overtakes gold diff by 25 min** (r=0.53) - **CS Diff** plateaus around r=0.20 - farming differences alone don't scale in importance - **Kills** grow steadily but remain the weakest individual predictor - Sweet spot for prediction: **15-minute mark** offers a good balance of signal vs "game still undecided" - XP overtaking gold at 25 min is interesting - suggests level/scaling advantages compound in mid-late game

Plot 4: Early-Game Correlation Heatmap (`04_early_game_correlation_heatmap.png`)

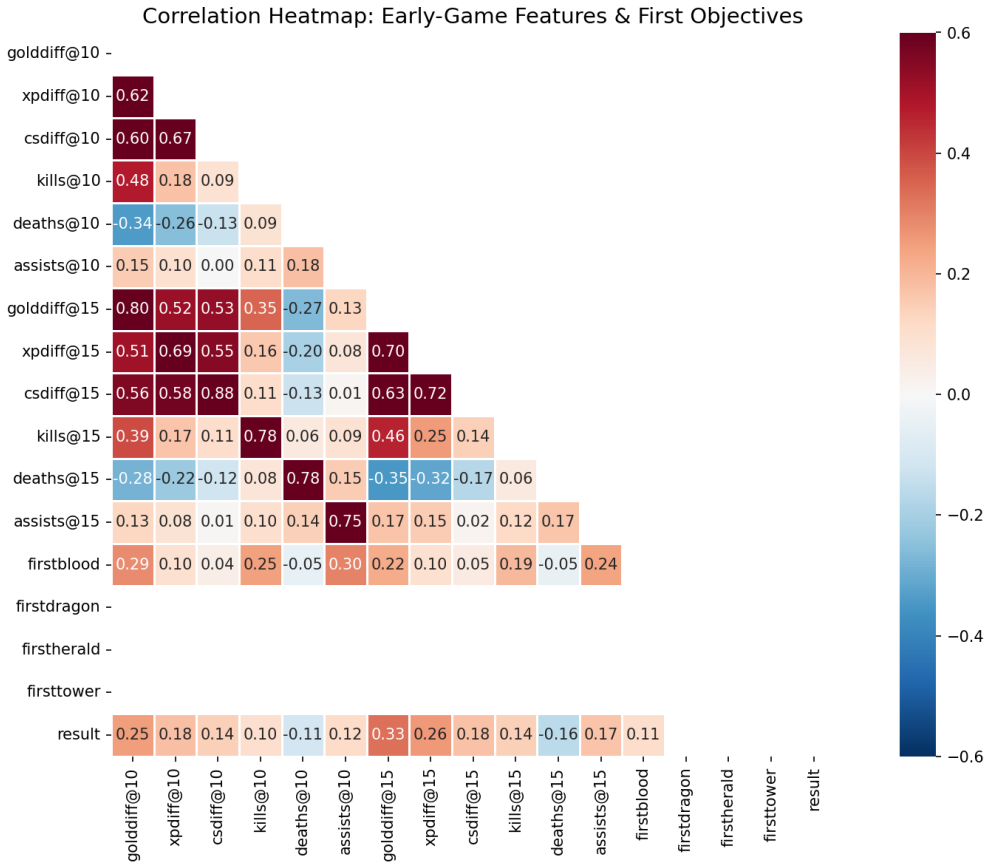

Figure 4: Correlation Heatmap

Key insights: - Gold diff, XP diff, and CS diff at same timestamps are moderately correlated (r~0.55-0.70) but NOT redundant - each carries independent signal - **golddiffat15** has the highest single-feature correlation with result (r=0.34) - First objectives (firstblood, firstdragon, firstherald, firsttower) have **low mutual correlations** - they are largely independent events - First tower correlates most with result among the "first" objectives (r=0.24 from heatmap context) - Kills and assists at timestamps are intercorrelated - feature selection should handle this - Deaths at 10/15 show negative correlation with win, as expected

Plot 5: Objective Count vs Win Rate (`05_objectives_vs_winrate.png`)

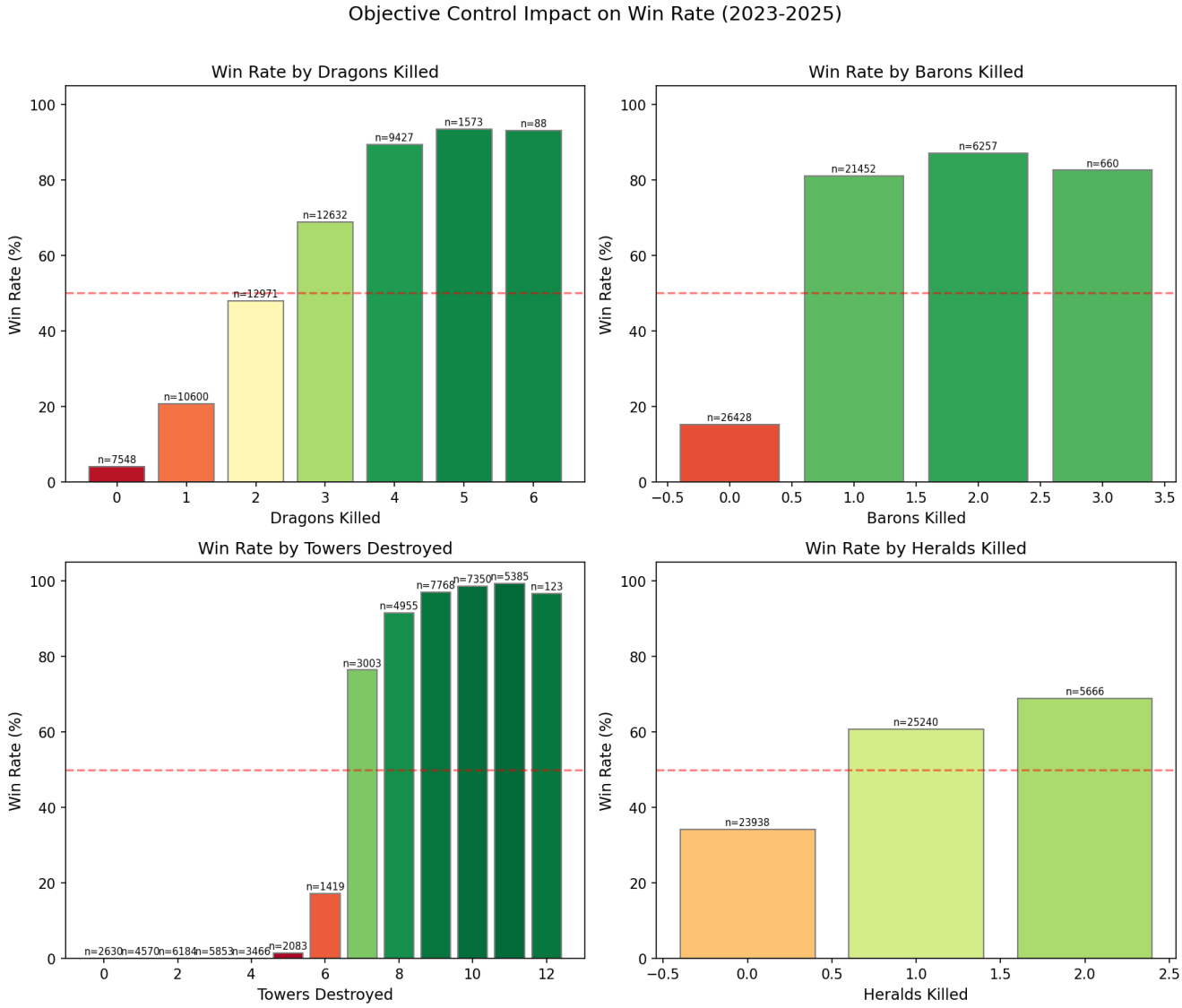

Figure 5: Objectives vs Win Rate

Key insights: - **Towers** show the clearest monotonic relationship: 0-5 towers = <1% win rate, 6 towers = 15%, 11+ towers = 98% - **Dragons**: 0 dragons = 5% win rate, 4+ dragons (soul) = 90%+ win rate. The "dragon soul" threshold (4 drakes) is clearly visible - **Barons**: binary impact - 0 barons = 15% win rate, 1+ baron = 80%+. Getting baron is nearly decisive - **Heralds**: 0 heralds = 34%, 1 herald = 61%, 2 heralds = 69%. Weaker effect but still meaningful - These are post-game stats (data leakage territory), but useful for understanding the game's mechanics and for the Shiny app visualization

Plot 6: Role-Specific Gold Diff Importance (`06_role_golddiff_correlation.png`)

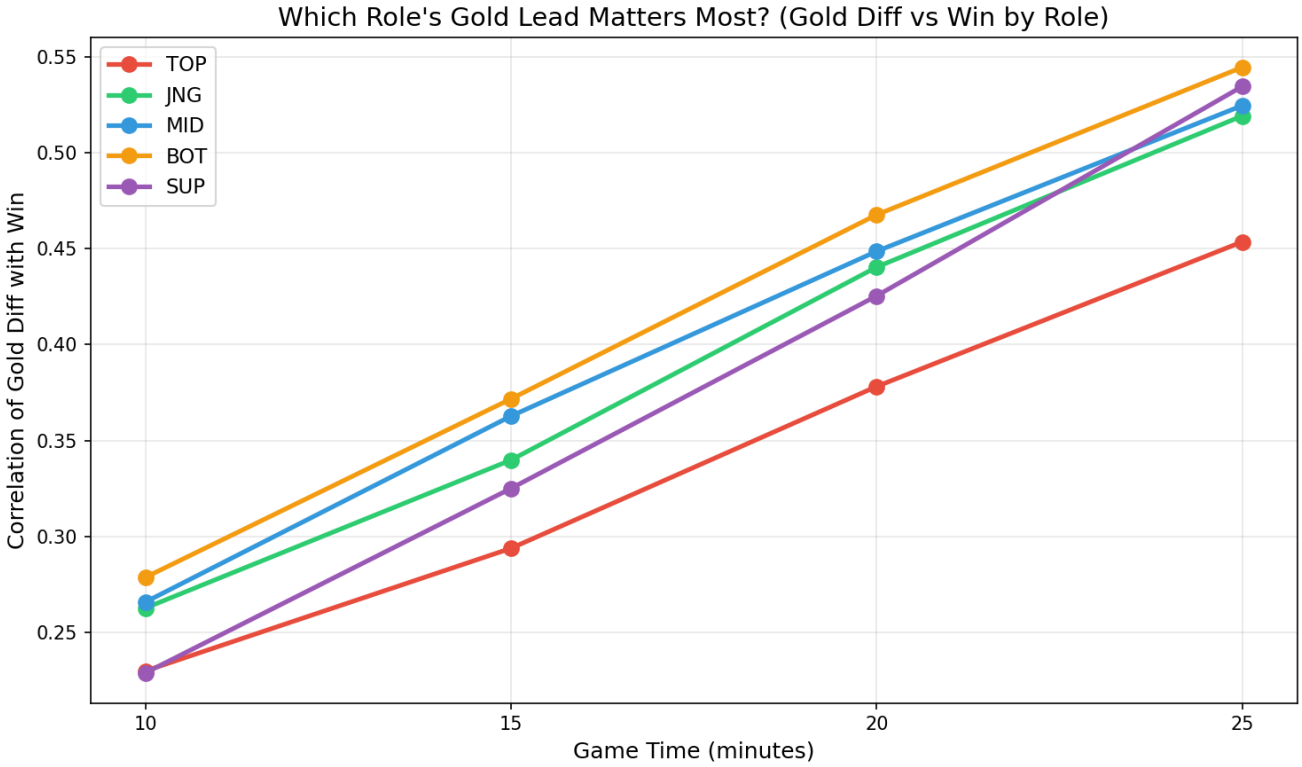

Figure 6: Role Gold Diff Correlation

Key insights: - **BOT (ADC)** gold lead is most predictive early (r=0.28 at 10min) - feeding the carry matters most early - **TOP** is consistently the least predictive role for gold diff - top lane is an "island" - **SUP** starts weakest at 10min but **surges to highest correlation by 25min** (r=0.54) - support gold scaling becomes crucial in mid-game - **MID and JNG** track closely together in the middle - All roles converge toward r=0.45-0.55 by 25 min, but the **early divergence is significant** for feature engineering - Implication: role-weighted features (weighting BOT gold diff higher at 15min) could improve early prediction

Plot 7: Blue Side Advantage by League (`07_blue_side_advantage.png`)

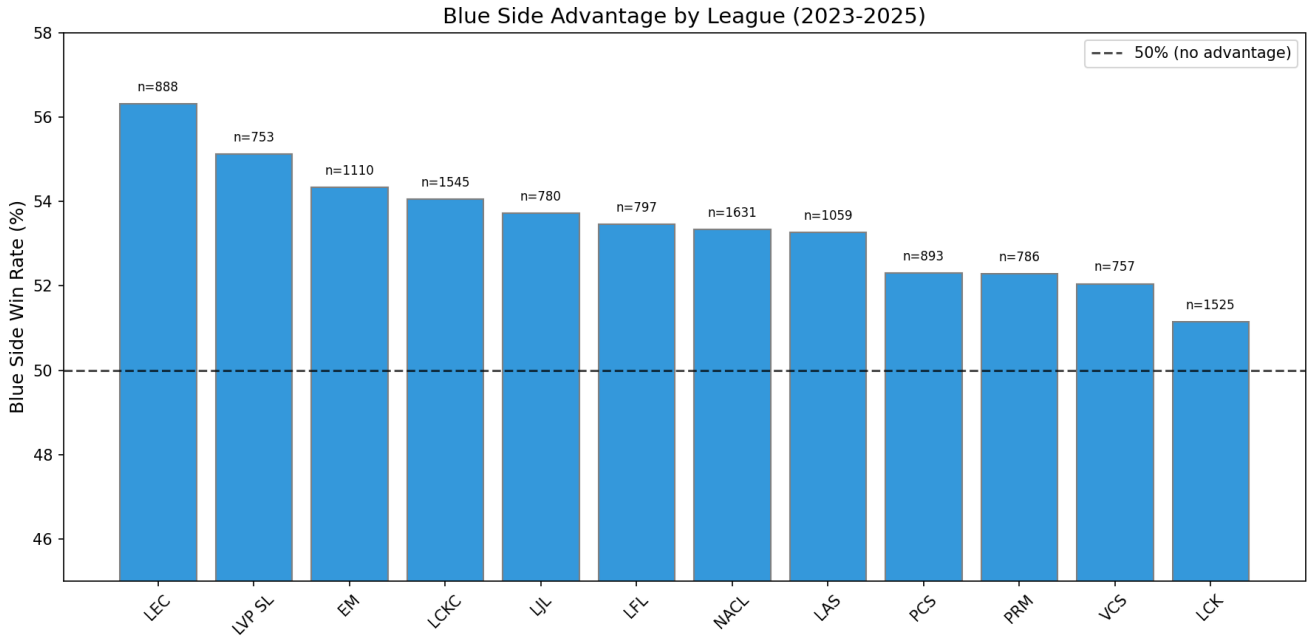

Figure 7: Blue Side Advantage

Key insights: - Blue side wins more than 50% across **every single league** - consistent structural advantage - **LEC** has the strongest blue side advantage at 56.4% - nearly 13% absolute edge - **LCK** has the weakest at 51.1% - Korean teams are best at playing from red side - Average blue side advantage is ~53% across all leagues - This confirms `side` should be included as a feature in the model - The variation across leagues suggests potential for league-specific models or interaction features (side x league)

Plot 8: 2025 Objective Mechanics - Atakhan, Void Grubs, Dragons (`08_2025_objectives_atakhan_grubs.png`)

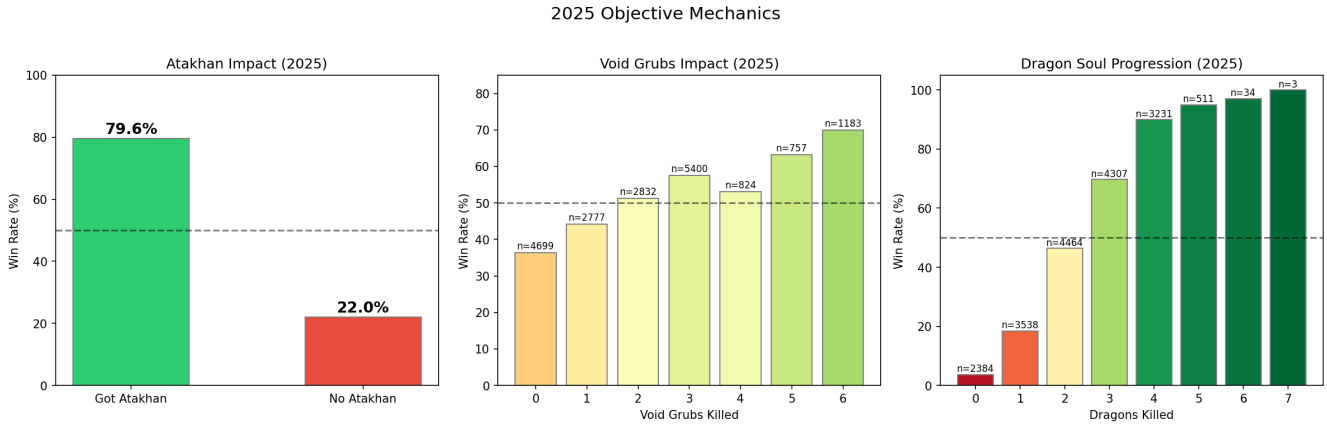

Figure 8: 2025 Objectives

Key insights: - **Atakhan** is a dominant binary predictor: 79.6% WR when secured vs 22.0% when not. Nearly as strong as Baron - **Void Grubs** show a clean monotonic relationship: 0 grubs = 36%, 3 grubs = 58%, 6 grubs = 70% - **Dragon Soul** threshold at 4 drakes is dramatic: 46% at 2 drakes jumps to 90% at 4 (soul) - All three objectives show strong, interpretable relationships - excellent features for modeling

Plot 9: Comeback Probability by Gold Deficit (`09_comeback_by_gold_deficit.png`)

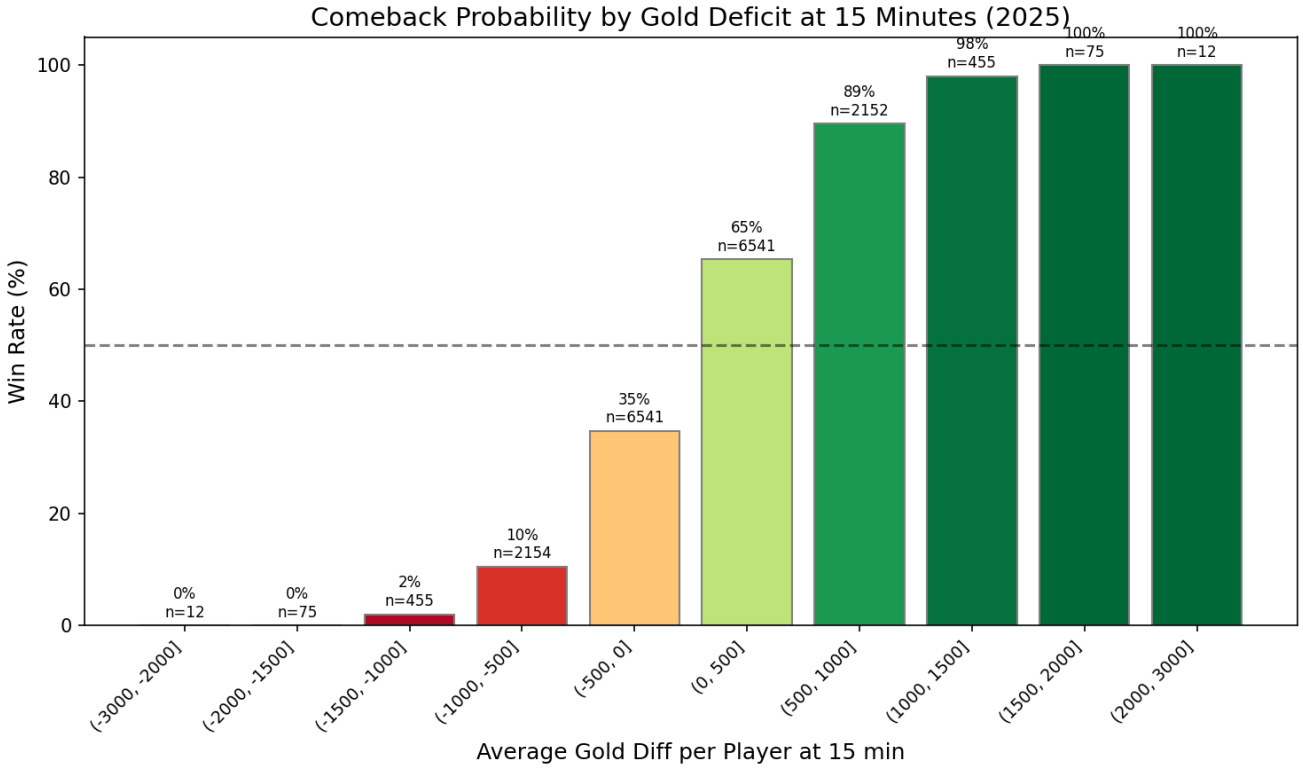

Figure 9: Comeback Analysis

Key insights: - **Extremely steep S-curve**: small leads (+500g) already give 65% WR, +1000g gives 89% - **Comebacks are nearly impossible** from large deficits: >1500g deficit = 0% win rate in 2025 - The -500 to +500 zone is where most games are contested (n=6,541 each side) - **2025 pro meta is highly snowball-dependent** - early leads compound ruthlessly - This validates using 15-min gold diff as the primary early-game predictor

Plot 10: Snowball Chains - First Objective Combinations (`10_snowball_chains.png`)

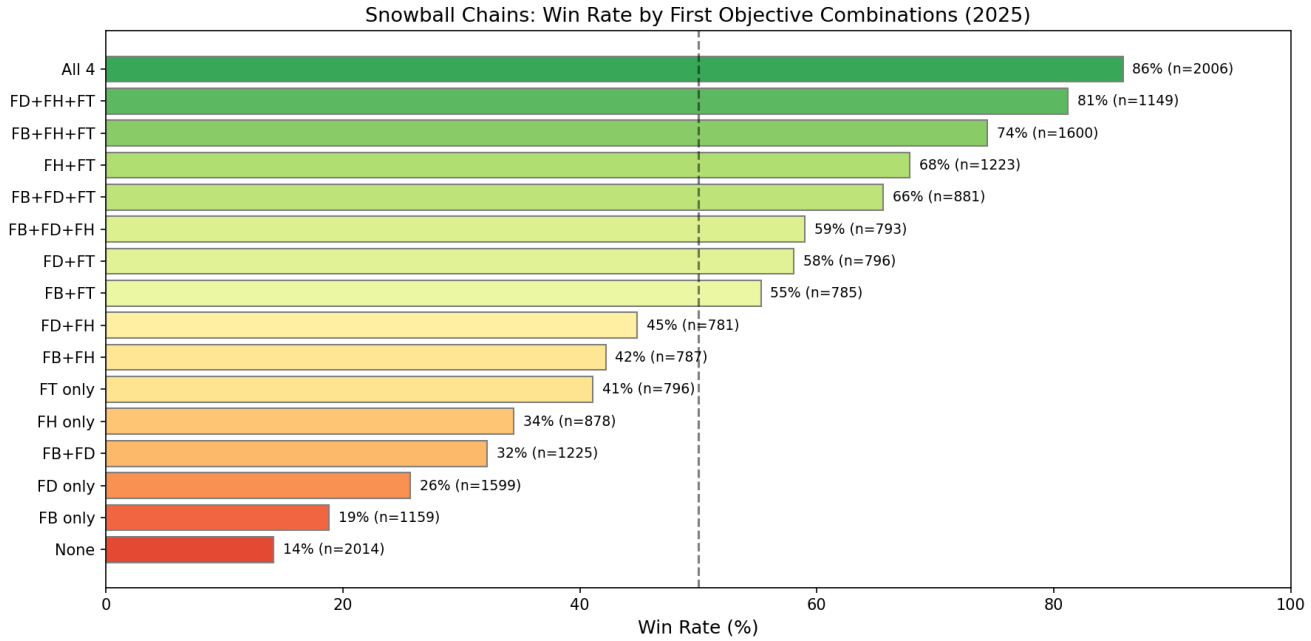

Figure 10: Snowball Chains

Key insights: - **All 4 firsts = 86% WR** - but diminishing returns beyond 3 objectives - **FH+FT (Herald + Tower) combo = 68%** is the most efficient 2-objective chain - **First Blood alone = only 19%** - barely better than getting nothing (14%) - First Dragon alone (26%) is also weak by itself - **Tower-related objectives dominate**: any combo including First Tower significantly boosts WR - Implication: objective interaction features (combinations) are more predictive than individual first-objective flags

Plot 11: First Blood by Role (`11_first_blood_by_role.png`)

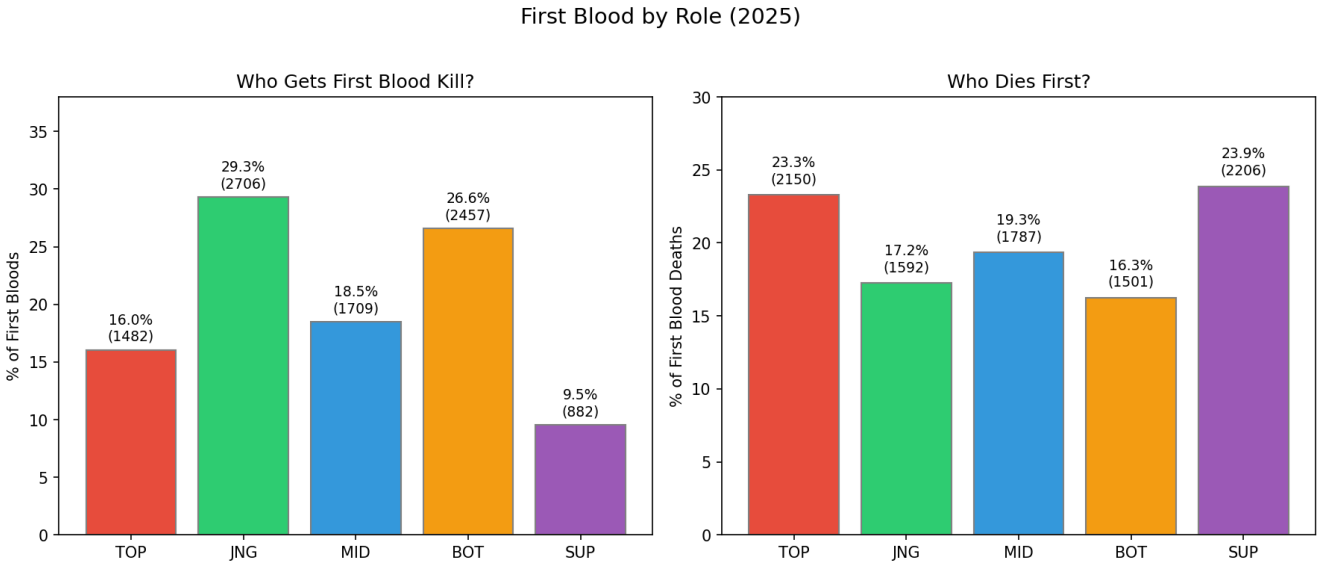

Figure 11: First Blood by Role

Key insights: - **Junglers are the primary first blood playmakers** (29.3%) - ganking is their core early-game function - **BOT (ADC)** is the second most common first blood killer (26.6%) - bot lane 2v2 fights - **Supports and top laners are the most common victims** (~24% each) - supports die roaming/warding, top laners die to ganks in isolation - **BOT (ADC) is the least likely to die first** (16.3%) despite being in a 2v2 lane - protected by support - Role-specific first blood patterns could be useful for role-weighted prediction models

Plot 12: Game Pace by League (`12_game_pace_by_league.png`)

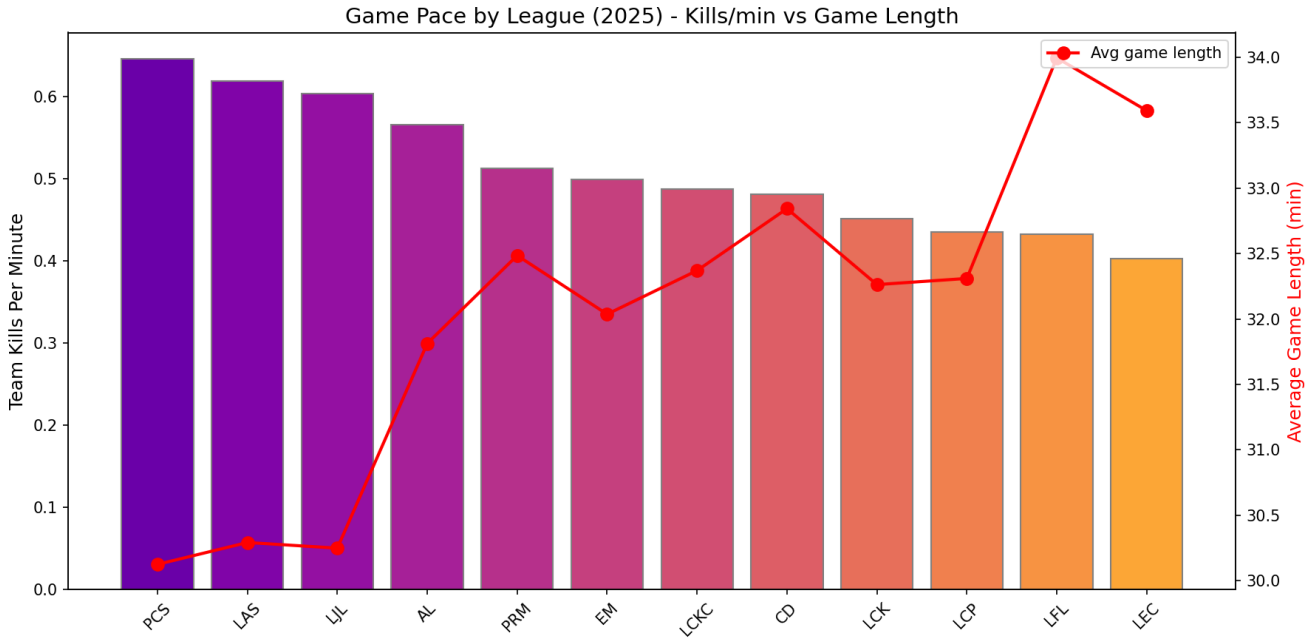

Figure 12: Game Pace by League

Key insights: - **Inverse relationship**: high kill leagues (PCS, LAS, LJL) have shorter games; low kill leagues (LEC, LFL) have longer games - **PCS** is the bloodiest league: 0.65 kills/min, ~30 min avg - **LEC** is the most controlled: 0.40 kills/min, ~34 min avg - 63% slower pace - This confirms **league as a critical contextual feature** - a "good" early gold lead means different things in PCS vs LEC - Models should either include league as a feature or be trained per-league for best results

---

Possible Hypothesis Formulations

Based on the 2025 deep dive, here are concrete hypothesis options ranked by strength:

H1 (Recommended): "Early-game state predicts match outcome"

"Can team-level early-game metrics at 15 minutes (gold diff, XP diff, first objectives, first blood) predict the winner of a professional LoL match?" - Baseline: simple gold-lead rule = 72.9% accuracy - Target: beat this with ML models using combined early-game features - Rich feature space: gold/xp/cs diffs + 4 first-objective flags + kills/deaths + side + void grubs - Clean classification problem, no data leakage

H2: "Objective priority strategy matters more than combat"

"Do structural objectives (towers, heralds, dragon control) predict victory better than combat metrics (kills, damage) in the early game?" - Compare feature groups: combat features vs objective features - Supported by data: first tower (r=0.40) » first blood (r=0.15) - Could use feature selection to prove which group is more important

H3: "Regional playstyle affects prediction models"

"Do early-game prediction models perform differently across leagues with different playstyles (aggressive PCS vs controlled LEC)?" - Train a model on LCK, test on PCS - does it transfer? - Compare model accuracy per league - Explore whether league-specific models outperform global models

H4: "New 2025 mechanics (Atakhan, Void Grubs) changed the game"

"How much do 2025-specific objectives (Atakhan, Void Grubs) contribute to win prediction beyond traditional objectives?" - Compare model with vs without Atakhan/void_grubs features - Atakhan has r=0.576 correlation - stronger than dragons (0.595) but binary

H5: "The snowball effect - when is the game decided?"

"At what point in the game does the early-game state become decisive, and what is the minimum combination of early advantages needed to predict victory with >80% confidence?" - Analyze prediction accuracy at 10, 15, 20, 25 min - Find the "tipping point" where comebacks become impossible (~1000g lead) - Combine with first-objective chains